NANYANG TECHNOLOGICAL UNIVERSITY

SEMESTER 1 EXAMINATION 2023-2024

EE7204 – LINEAR SYSTEMS

November / December 2023 Time Allowed: 3 hours

INSTRUCTIONS

1. This paper contains 5 questions and comprises 5 pages.
2. Answer all 5 questions.
3. All questions carry equal marks.
4. This is a closed book examination.
5. Unless specifically stated, all symbols have their usual meanings.
6. Laplace–Transform Table is included in Appendix A on page 5.

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1. Consider the linear time-invariant system

$$\dot{x}(t) = Ax(t) + Bu(t), \quad y(t) = Cx(t)$$

where x is the state, u is the input, y is the output,

$$A = \begin{bmatrix} -2 & 1 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & -2 \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}.$$

(a) Find a linear transformation that transforms the matrix A into the diagonal or Jordan form and thus compute e^{At}.

(15 Marks)

(b) Given the initial state $x(0) = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ and the input $u(t) = 1, t \geq 0$, find the output of the system $y(t)$.

(5 Marks)

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2. Consider the linear time-invariant system

$$\dot{x}(t) = Ax(t) + Bu(t),$$

where x is the state, u is the input,

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 0 & -1 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 0 \\ \alpha \end{bmatrix}$$

with α being a constant.

(a) Determine the controllability of the system with respect to the parameter α.

(8 Marks)

(b) For the values of α found in Q2(a), if any, such that the system is uncontrollable, find their corresponding controllable canonical decomposition of the system and determine if the system is stabilizable.

(12 Marks)

3. (a) Consider the linear time-invariant system

$$\dot{x}(t) = Ax(t) + Bu(t), \quad y(t) = Cx(t)$$

where x is the state, u is the input, y is the output,

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad C = \begin{bmatrix} -1 & 1 & 0 \end{bmatrix}.$$

Determine if the system is internally stable and bounded-input and bounded-output stable.

(10 Marks)

(b) Given the following polynomial matrix

$$P(s) = \begin{bmatrix} s+1 & s+2 \\ s^2+6s+9 & s^2+4s+3 \\ s+3 & s^2+5s+6 \end{bmatrix},$$

determine its rank and all possible value(s) of s that could reduce its rank.

(10 Marks)

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4. Given a strictly proper transition $G(s)$ below, answer the following questions.

$$G(s) = \begin{bmatrix} \dfrac{s(s+1)}{(s+2)^2(s^3+4s^2+5s+2)} & \dfrac{s}{(s+2)^2} \\ \dfrac{-s(s+1)-s^2(s+2)^2}{(s+2)^2(s^3+4s^2+5s+2)} & \dfrac{-s}{(s+2)^2} \end{bmatrix}$$

(a) Determine a Right Polynomial Fraction Description (RPFD) of $G(s)$, where $N(s)$ is specifically chosen as

$$N(s) = \begin{bmatrix} s & 0 \\ -s & s^2 \end{bmatrix}.$$

(5 Marks)

(b) Determine whether the calculated RPFD in (a) is a coprime RPFD (CRFPD). If not, find a CRPFD of $G(s)$.

(6 Marks)

(c) Determine whether the CRPFD obtained in (b) is column reduced. If not, find a column reduced CRPFD of $G(s)$.

(6 Marks)

(d) Determine the McMillan degree of $G(s)$ and decide the minimum dimension of any realization of $G(s)$.

(3 Marks)

5. Given the following block diagram in Figure 1, where $G(s) = \frac{s-1}{s^2-2}$ is the transfer function of the plant, w and r are external disturbance and reference signal, respectively, answer the following questions:

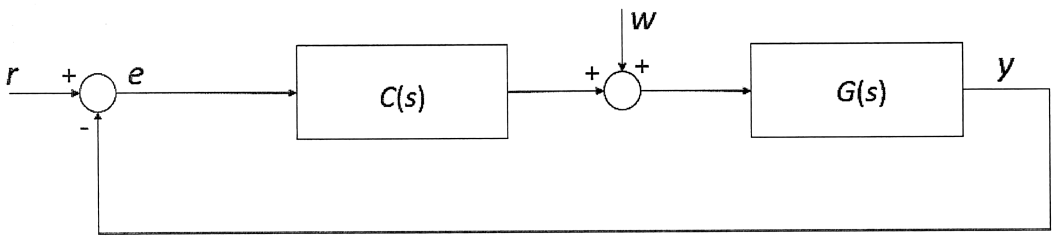

Figure 1

Note: Question No. 5 continues on page 4

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(a) Assume that there is no disturbance, namely $w(t) = 0$. Design a compensator $C(s)$ such that the transfer function of the closed-loop system has five stable poles at -1, -1, -1, -1, -1, and the system can track any step reference signal asymptotically.

(14 Marks)

(b) Assume that the disturbance signal $w(t)$ is an arbitrary step function. Design a compensator $C(s)$ such that the transfer function of the closed-loop system has five stable poles at -1, -1, -1, -1, -1, and the system can robustly track any step reference signal asymptotically and reject the impact of $w(t)$ asymptotically.

(6 Marks)

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APPENDIX A

Laplace-Transform Table

LAPLACE TRANSOFRM $F(s)$	TIME FUNCTION $f(t)$
$\frac{1}{s}$	Unit-step function $u(t)$
$\frac{1}{s+\alpha}$	$e^{-\alpha t}$
$\frac{\alpha}{s(s+\alpha)}$	$1-e^{-\alpha t}$
$\frac{1}{(s+\alpha)^2}$	$te^{-\alpha t}$
$\frac{a}{s^2+a^2}$	$\sin at$
$\frac{s}{s^2+a^2}$	$\cos at$
$\frac{a}{(s+b)^2+a^2}$	$e^{-bt}\sin at$
$\frac{s+b}{(s+b)^2+a^2}$	$e^{-bt}\cos at$
$\frac{\omega_n^2}{s(s^2+2\zeta\omega_n s+\omega_n^2)}$	$1-\frac{1}{\sqrt{1-\zeta^2}}e^{-\zeta\omega_n t}\sin(\omega_n\sqrt{1-\zeta^2}\,t+\theta)$ where $\theta=\cos^{-1}\zeta \qquad (\zeta<1)$
$\frac{\alpha^2}{s(s+\alpha)^2}$	$1-e^{-\alpha t}(1+\alpha t)$

END OF PAPER

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Please read the following instructions carefully:

1. **Please do not turn over the question paper until you are told to do so. Disciplinary action may be taken against you if you do so.**

2. You are not allowed to leave the examination hall unless accompanied by an invigilator. You may raise your hand if you need to communicate with the invigilator.

3. Please write your Matriculation Number on the front of the answer book.

4. Please indicate clearly in the answer book (at the appropriate place) if you are continuing the answer to a question elsewhere in the book.

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